The Decision Value of Hidden Supply-Network Structure: Zero-Value Conditions and Identification-Premium Bounds

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Abstract

Firms invest in sub-tier supply-network visibility through mapping, supplier questionnaires and audits, and a substantial literature studies how hidden supplier structure can be inferred. We study a prior question: what is that structure worth to a specific mitigation decision? We formulate discovery-coupled mitigation as a probe-then-secure problem, with no probe cost priced here, on a partially observed supply directed acyclic graph and analyse the expected value of perfect structural information (EVPI). We give sufficient structural conditions, namely invariance of the loss distribution under identification-free plans together with uniform plan-level dominance of identification-free actions, under which EVPI is exactly zero; since perfect information dominates any acquisition policy, every feasible information-acquisition policy then has zero decision value within the modelled problem. The conditions are stated for any monotone law-invariant objective, so they cover expected loss and conditional value at risk alike. The first condition holds when a symmetry of the public network exchanges the candidate suppliers of each category together with their consumers, upstream subtrees and correlation clusters; exchangeability of the candidates' marginal risks alone does not suffice, and we give a counterexample. For the second we give a sufficient condition read off the graph and the cost vector. Outside the zero-value regime we bound EVPI above by a sum of per-category identification premiums computable from prior and public data, prove the bound for node-disjoint categories, give the premium exactly in a dilution special case, and identify a sub-case in which the bound is attained. Exact enumeration of synthetic miniature instances found no violation of any stated result. The conditions and the bound are not evaluated on real supply networks here.

Keywords: value of information, EVPI, supply network visibility, disruption risk, conditional value at risk, endogenous uncertainty

1 Introduction

Sub-tier supply-network visibility is bought through mapping services, supplier questionnaires, audit programmes and related discovery activities. The case for such spending is usually argued from the operational consequences of incomplete visibility. It is much less common to price the information itself: to ask by how much a firm's mitigation decisions would improve if the hidden structure were supplied at no cost.

Decision analysis answers that question with the expected value of perfect information, which bounds the improvement obtainable from any amount of investigation. Applying it to supply-network structure is easy to state and harder to use, because the decision the information feeds — which suppliers to qualify a backup for, under a cost budget, on a network whose upstream shape is the unknown — is combinatorial.

We formulate this problem as a decision model of discovery-coupled mitigation and characterise its information value. One point of scope should be clear from the outset: we value the information, not the coupling. No probe cost and no shared probe-and-mitigation budget enters any quantity studied here, since the uninformed planner acts on the prior and the informed planner is handed θ outright. The first result is negative and, in our view, the more useful one: there are structural conditions under which the value of perfect structural information is exactly zero. Both can be examined before any structural discovery is commissioned: the first, a symmetry of the public network that exchanges the candidates of each category together with their consumers, upstream subtrees and correlation clusters, directly from public data, and the second through a sufficient condition on the graph and the cost vector. The result is independent of the optimisation algorithm used to implement the decision: under those conditions no information-acquisition policy can improve the objective, because perfect information dominates every such policy and is itself worth nothing. The second result is a bound for the cases where those conditions fail, expressed as a sum of per-category identification premiums, each computable from information the firm already holds.

Central claim.

We characterise sufficient structural conditions under which the decision value of hidden supplier information is exactly zero, and derive a computable upper bound on that value outside the zero-value regime. Everything else in the paper supports that claim: the enumeration experiments are verification, the CVaR statement is an extension, and the shared-subtree case is an unresolved extension we label as such.

Contributions.

1. *Zero-value characterisation.* Theorem 4 gives sufficient conditions for EVPI $= 0$, and Lemma 2 transfers the conclusion to every information-acquisition policy. Lemma 13 gives a sufficient condition for the first hypothesis, a symmetry of the public structure, and Example 1 shows that exchangeability of the candidates' marginal risks does not suffice. Proposition 8 gives a sufficient condition for the harder of the two hypotheses that is read off the public graph and cost vector. Corollary 5 extends the statement to monotone law-invariant objectives, covering conditional value at risk.

2. *Identification-premium upper bound.* Theorem 6 bounds EVPI by a finite, directly computable sum of per-category premiums for node-disjoint categories, and Corollary 9 bounds the partial value of each category separately. Computable EVPI bounds are not new (Section 2, stream A′). What we believe to be new, and have not found stated elsewhere, is a bound of this form: it requires no convexity, decomposes over a discrete latent assignment, and is localised to individual channels: under the assumptions of Theorem 6, a channel whose premium is zero has zero partial value by Corollary 9. Whether ranking channels by premium identifies those worth discovering is not established here. Further, Corollary 7 gives the exact premium in a dilution special case, and we identify a sub-case of it in which the bound is attained with equality; and Proposition 10 gives the premium in closed form under a specified dispersion family, identifying a threshold below which dispersion does not enter the bound at all.
3. *Computational verification.* Exhaustive enumeration of exact miniature instances tests the implementation of the definitions, compares the per-channel premium with exact EVPPI, and searches for violations of the stated results and for counterexamples to Conjecture 12 (Section 6). It is evidence about small synthetic instances, not a measurement on supply networks.

Section 7 states what the paper does not establish.

2 Related work and positioning

(A) Value of information.

The expected value of perfect information originates with Raiffa and Schlaifer [22] and Howard [15]. Its partial form, EVPPI, is standard in health economics, where the computational cost of nested estimation is a recognised difficulty [14, 26]. Our premium is an EVPPI-shaped quantity for structural parameters of a network, evaluated by finite enumeration over budget breakpoints rather than by nested simulation. We claim no priority over that literature, whose machinery we reuse.

(A′) Bounding EVPI when it cannot be computed.

We must be explicit that a computable upper bound on EVPI is not a new idea. Because exact evaluation is often prohibitive, bounds have been derived since Madansky [21], and Avriel and Williams [5] obtained EVPI bounds evaluable as part of a post-optimal analysis of a linear or convex program. Ziemba and Butterworth [30] and Hausch and Ziemba [13] devote a two-part study to bounding the value of information directly; Huang, Vertinsky and Ziemba [16] sharpen the perfect-information bounds for simple-recourse problems; and Birge [6] develops the companion bounds for the value of the stochastic solution. That family of bounds rests on convexity or concavity: Huang, Vertinsky and Ziemba state theirs for simple-recourse problems with nonlinear return functions and concave utility functions, using extensions of the Jensen and Edmundson–Madansky inequalities. Theorem 6 uses no convexity and no continuity: it is a combinatorial decomposition over a discrete latent assignment, evaluated on a finite set of budget breakpoints, and it localises the bound to individual categories, so

that a category whose premium is zero is shown to have zero partial value. Our contribution is that specific form, not the general idea of bounding EVPI, and where the two apply to the same problem we make no claim that ours is tighter.

(B) Information acquisition and active learning.

Adaptive information gathering with performance guarantees is well developed [12], and decision-coupled acquisition, which drives uncertainty into a single decision region instead of reducing it uniformly, is the closest precedent for the objective used here [17]. The ordering of information structures by their decision value is Blackwell's [7], and Lemma 2 is a standard consequence of it rather than a contribution of this paper. That literature asks how to acquire information efficiently. We ask when acquisition cannot help at all, which is logically prior and is not settled by any performance guarantee.

(C) Decision-dependent information discovery.

Stochastic programming with endogenous uncertainty studies problems in which decisions determine what is revealed [11, 18], and the robust counterpart optimises under decision-dependent revelation [28]. Artstein and Wets [1] and Artstein [2, 3] treat the information structure itself as a decision variable through the notion of sensors; Artstein [2] recasts a two-stage problem with a probing decision as a three-stage stochastic program and establishes a strong law of large numbers for it, and Artstein [3] addresses the trade-off between the gain from new information and the cost of obtaining it. Closest to our setting is the probing-enhanced stochastic program of Ma et al. [20], in which costly discrete probes are chosen before a two-stage decision and the perfect-information bound is the benchmark to be improved upon; their contribution is algorithmic, a branch-and-bound scheme over probing sets. The direction of the question differs: that stream computes policies *under* revelation, while we characterise conditions in which revelation changes nothing, so that the problem those methods solve has a degenerate answer.

(D) Valuation of supply-chain visibility.

It would be wrong to claim that visibility has not been valued. Caridi et al. [9] develop a structured value-assessment model for visibility and apply it to a real chain, and recent student and practitioner work continues to estimate returns to deeper tiers [10]. That work maps visibility improvements to operational and strategic benefits, typically through causal models of performance indicators. Our question is narrower and its answer sharper: for a specified mitigation decision and a specified loss, when is the decision value of latent structure exactly zero, and when it is not, what upper bound can be computed in advance. Neither statement subsumes the other.

(E) Multi-tier structure and hidden links.

Inference of hidden supplier relationships is an active area [8, 19], and topological measures of network vulnerability are well established [24, 29]. These methods find or score structure; they do not price it against a decision.

(F) Disruption mitigation.

Mitigation and contingency strategy follows Tomlin [27], with the wider OR/MS literature surveyed by Snyder et al. [25]. Our mitigation model — budgeted backup qualification under a conjunctive cascade — is deliberately simple, since the object of study is the information rather than the mitigation policy.

(G) Risk measures.

The tail machinery is that of Rockafellar and Uryasev [23] and Artzner et al. [4].

Table 1 summarises where this paper sits against these streams. The column that separates it is the third.

Table 1 Positioning against adjacent streams. The column that separates this paper is the sufficient zero-value condition; computable EVPI bounds already exist (stream A′), and our claim there concerns the form of the bound rather than its existence. The zero-value column records what we found in the works we examined, not an exhaustive claim about each stream. "Model only" means that our results are stated for a model of a supply network and are not evaluated on real networks.

stream	values latent structure	gives a zero-value condition	gives an EVPI bound	bound is convexity-free	applied to supply networks
Value of information (A)	yes	no	no	n/a	no
EVPI bounding (A′)	no	no	yes	not in those examined	no
Active acquisition (B)	yes	no	no	n/a	no
Decision-dependent discovery (C)	yes	no	no	n/a	no
Visibility valuation (D)	yes	no	no	n/a	yes
Hidden-link inference (E)	no	no	no	n/a	yes
This paper	yes	yes	yes	yes	model only

3 Model

Network.

A finite directed acyclic graph $G = (V, E)$, where $u \to v$ means u supplies v. A focal node 0 consumes K_0 input categories. Every node v consumes categories $k \in K_v$, each served by a set of suppliers. Service is conjunctive across categories and substitutable within a category, in the sense that any one working supplier suffices. That clause fixes the semantics of the cascade rather than the sourcing structure. The candidate set is epistemic rather than a redundancy: exactly one candidate actually supplies the category, and the set records who it might be. The model therefore contains no concurrently multi-sourced category, and dual sourcing as a mitigation lies outside it. Each node has a publicly known securing cost c_v and belongs to a publicly known correlation cluster. A node's *tier* is its distance from the focal node along the supply

direction: the focal firm's direct suppliers are tier one, their suppliers tier two, and so on.

Uncertainty.

The structure parameter θ specifies, for each latent category (v, k), which candidate in a finite set $C_{v,k}$ is the true supplier, exactly one candidate serving the category, an assumption Corollary 7 and Proposition 10 use and, in the dispersion family of Definition 2, which candidate carries which risk attribute. A prior π over θ is given. Probing a node reveals its candidate in-edges through a noisy asymmetric channel. That channel belongs to the wider model, of which this paper studies only the value of the information a probe could deliver; it enters no quantity studied here, since the uninformed planner sees only the prior and the informed planner sees θ outright.

Disruption and loss.

A scenario ω draws node-level failures, correlated within clusters by a one-factor Gaussian copula. Operability propagates through the conjunctive cascade. A plan $S \subseteq V \setminus \{0\}$ with $c(S) \leq B$ qualifies backup sources: $u \in S$ guarantees u's output to its consumers. The focal loss is the fraction of focal categories starved,

$$L(S; \theta, \omega) = \frac{1}{K_0} \sum_{k=1}^{K_0} Z_k, \qquad Z_k = \mathbf{1}\{\text{focal category } k \text{ starved}\}. \tag{1}$$

Objectives and the exact target.

Write $j(S; \theta) = \mathbb{E}_\omega L(S; \theta, \omega)$ and $J(S; \theta) = \mathrm{CVaR}_{\alpha,\omega}(L(S; \theta, \omega))$. Let S_π minimise $\mathbb{E}_{\theta\sim\pi} j(S; \theta)$ and S_θ minimise $j(S; \theta)$. Then

$$\mathrm{EVPI}(\pi) = \mathbb{E}_\pi[j(S_\pi; \theta)] - \mathbb{E}_\pi[j(S_\theta; \theta)] \;\geq\; 0. \tag{2}$$

Units.

L is a fraction of focal categories, so every quantity here is dimensionless. A value of 0.005 for EVPI or for the bound below means that perfect structural knowledge would reduce the objective by half a percentage point of the focal basket: expected category starvation where the objective is j, and the conditional tail average of the starved fraction where it is J. Because L is normalised by K_0, rescuing one focal category moves the objective by $1/K_0$, so the same absolute value means different things on networks with different numbers of focal categories, and K_0 times a value counts the focal categories rescued. Converting either into money requires the cost of one starved category, which the model does not represent.

Channels.

A *channel* is one latent category at one node, and is the unit at which identification premiums are computed. A *focal category* is one of the K_0 inputs the focal node consumes.

3.1 What an implemented planner observes

A firm operates an implemented planner rather than an exact optimiser, and what it can observe by running that planner with and without θ is not (2). Let $\hat{S}_\pi$ and $\hat{S}_\theta$ be the plans returned by the implemented planner given the prior and given θ. Define the *heuristic oracle advantage*

$$H \;=\; \mathbb{E}_\pi\Big[j(\hat{S}_\pi;\theta)\Big] - \mathbb{E}_\pi\Big[j(\hat{S}_\theta;\theta)\Big]\,. \tag{3}$$

Proposition 1 (Exact relation between H and EVPI) *Let $\varepsilon_\pi = \mathbb{E}_\pi[j(\hat{S}_\pi;\theta)] - \mathbb{E}_\pi[j(S_\pi;\theta)] \geq 0$ and $\varepsilon_\theta = \mathbb{E}_\pi[j(\hat{S}_\theta;\theta) - j(S_\theta;\theta)] \geq 0$ be the planner's optimality gaps in the two regimes. Then*

$$H = \mathrm{EVPI}(\pi) + \varepsilon_\pi - \varepsilon_\theta. \tag{4}$$

Hence $H \leq \mathrm{EVPI}$ if and only if $\varepsilon_\pi \leq \varepsilon_\theta$. In general the sign of $\varepsilon_\pi - \varepsilon_\theta$ is not determined, so H is neither an upper nor a lower bound on EVPI; if the planner is θ-optimal when given θ, so that $\varepsilon_\theta = 0$, then $H \geq \mathrm{EVPI}$.

Proof (E1) By definition of ε_π, $\mathbb{E}[j(\hat{S}_\pi;\theta)] = \mathbb{E}[j(S_\pi;\theta)] + \varepsilon_\pi$.
(E2) By definition of ε_θ, $\mathbb{E}[j(\hat{S}_\theta;\theta)] = \mathbb{E}[j(S_\theta;\theta)] + \varepsilon_\theta$.
(E3) Subtracting (E2) from (E1) and using (2) gives (4). Both gaps are nonnegative because S_π and S_θ minimise over the same feasible sets that the planner searches heuristically, and neither gap dominates the other in general, which gives the remaining statements. □

Proposition 1 holds for any fixed objective used in every term; under the CVaR objective J it relates H to the CVaR value of perfect information. A planner's observed advantage is therefore a planner-specific quantity, and it bounds EVPI from neither side unless the sign of $\varepsilon_\pi - \varepsilon_\theta$ is known.

4 When structural information has zero value

Lemma 2 (Acquisition is dominated by perfect information) *For any information-acquisition policy φ with value* $\mathrm{Val}(\varphi)$, $\mathrm{Val}(\varphi) \leq \mathrm{EVPI}(\pi)$.

Proof (L1) By definition, $\mathrm{Val}(\varphi) = \mathbb{E}_\pi[j(S_\pi;\theta)] - \mathbb{E}_o\mathbb{E}_{\theta|o}[j(S_{\pi_o};\theta)]$, where o is the observation induced by φ, π_o the posterior and S_{π_o} the plan chosen after o.
(L2) Pointwise in (o,θ), S_{π_o} is one feasible plan among others, so $j(S_{\pi_o};\theta) \geq \min_S j(S;\theta) = j(S_\theta;\theta)$.
(L3) Taking $\mathbb{E}_{\theta|o}$ of (L2), then $\mathbb{E}_o$, and applying the tower property gives $\mathbb{E}_o\mathbb{E}_{\theta|o}[j(S_{\pi_o};\theta)] \geq \mathbb{E}_\theta[j(S_\theta;\theta)]$.
(L4) Substituting (L3) into (L1) yields $\mathrm{Val}(\varphi) \leq \mathrm{EVPI}(\pi)$. □

Proposition 3 (Bernoulli CVaR) *For $Z \in \{0,1\}$ with $\mathbb{P}(Z=1) = p$,* $\mathrm{CVaR}_\alpha(Z) = \min\{1, p/(1-\alpha)\}$, *and* $\mathrm{CVaR}_\alpha(K_0^{-1}\sum_k Z_k) \leq K_0^{-1}\sum_k \min\{1, p_k/(1-\alpha)\}$.

Proof (P1) If $p \geq 1-\alpha$ the worst $(1-\alpha)$ tail consists of ones only, so the tail average is 1.
(P2) If $p < 1-\alpha$ the tail contains all p mass of ones and $(1-\alpha-p)$ of zeros, so its average is $p/(1-\alpha)$.
(P3) Combining gives $\mathrm{CVaR}_\alpha(Z) = \min\{1, p/(1-\alpha)\}$.
(P4) CVaR is subadditive and positively homogeneous [4], so $\mathrm{CVaR}_\alpha(K_0^{-1}\sum_k Z_k) \leq K_0^{-1}\sum_k \mathrm{CVaR}_\alpha(Z_k)$, and (P3) gives the stated form. □

Definition 1 (Identification-free actions) An action is *identification-free* for the latent category (v,k) if the conditional law of the focal loss L, given a plan containing that action, does not depend on $\theta_{v,k}$. A plan is *identification-free* if each of its elements is identification-free for the category it addresses. This is a condition on the plan's own elements only: it says nothing about the latent categories the plan leaves untouched, whose θ-dependence is what (H1) below rules out. Canonical examples of identification-free actions: secure the consumer v; secure every candidate in $C_{v,k}$; and, under the sufficient condition (H1′) below, take no action, since every structure automorphism fixes the empty plan. Under heterogeneous latent candidate risk the last of these is not identification-free, and exchangeability of the candidates' marginal laws alone does not make it so, since the law of L under inaction can depend on which candidate is the supplier. Securing one named candidate $u \in C_{v,k}$ is *identification-dependent*, since its effect depends on whether u is the supplier.

We use two hypotheses.

(H1) *Loss-law invariance.* For every identification-free plan S, the law of $L(S;\theta,\cdot)$ is the same for all θ. The hypothesis is not implied by Definition 1: an identification-free plan constrains only the categories its own elements address, and (H1) is exactly the requirement that the categories it leaves alone contribute a law that does not vary with θ. That is why the sufficient condition (H1′) below is stated over every latent category rather than over the plan's.

(H2) *Uniform plan-level dominance.* Every identification-dependent element u of any plan admits an identification-free replacement $a(u)$ with $c(a(u)) \leq c(u)$ and $j(S\setminus\{u\}\cup\{a(u)\};\theta) \leq j(S;\theta)$ for all θ. Securing every candidate in the category is identification-free but cannot serve as $a(u)$ under positive costs: the replacement set contains u, so it is never cheaper. Only an action at the consumer level, or outside the category, can witness the cost clause.

A checkable sufficient condition for (H1).

Exchangeability of the candidates of each category in marginal operability law, in securing cost and under the prior is *not* sufficient. Two candidates with the same failure law can differ in their other consumers, in their upstream subtrees or in their correlation cluster, and each of these differences can make the joint law of the starvation indicators, and with it the law of L, depend on which candidate is the supplier. Example 1 in Appendix A gives an instance in which the marginal condition holds, (H2) holds by Proposition 8, and EVPI > 0. What suffices is a symmetry of the public structure. Write q_v for the marginal failure probability of node v and $\kappa(v)$ for its cluster label. A *structure automorphism* is a bijection $\sigma : V \to V$, together with bijections $\sigma_v : K_v \to K_{\sigma v}$ and a bijection η of cluster labels, such that (A0) $\sigma(0) = 0$;

(A1) whenever (v,k) has public supplier s, $(\sigma v, \sigma_v k)$ has public supplier $\sigma(s)$; (A2) (v,k) is latent if and only if $(\sigma v, \sigma_v k)$ is, and then $C_{\sigma v,\sigma_v k} = \sigma(C_{v,k})$; (A3) $q_{\sigma v} = q_v$, $\kappa(\sigma v) = \eta(\kappa(v))$, and η preserves each cluster's copula parameter. Because σ preserves every supply and candidate relation, it carries each node's upstream subtree and consumers along with the node. It acts on states by $(\sigma\theta)_{\sigma v,\sigma_v k} = \sigma(\theta_{v,k})$ and on plans by $\sigma S = \{\sigma u : u \in S\}$.

(H1′) *Automorphic exchangeability.* For every identification-free plan S and all θ, θ' there is a structure automorphism σ with $\sigma\theta = \theta'$ and $\sigma S = S$.

Lemma 13 in Appendix A proves that (H1′) implies (H1); in fact the joint law of $(Z_1, \ldots, Z_{K_0})$ is then invariant in θ up to a fixed permutation of the focal categories. Neither the prior nor the cost vector enters, since (H1) concerns a fixed plan pointwise in θ. A local form is easier to check. It suffices that for every latent category (v,k) and all $u, u' \in C_{v,k}$ some structure automorphism maps u to u', fixes v and k, fixes pointwise every other candidate set and every other latent category's consumer, and fixes every identification-free plan. The typical witness swaps the upstream cones of two candidates when the cones are disjoint, isomorphic with matching failure probabilities and cluster labels, and joined to the rest of the network only by the edge into (v,k). A candidate that belongs to two candidate sets, or a candidate with a consumer that its rivals lack, defeats the local form.

Theorem 4 (Zero decision value of hidden structure) *Under (H1) and (H2),* $\mathrm{EVPI}(\pi) = 0$*; hence by Lemma 2 every feasible information-acquisition policy has zero decision value.*

Proof (T1) Fix θ. The feasible plan set is finite, so a θ-optimal plan S_θ exists.
(T2) Suppose S_θ contains an identification-dependent element u, and replace it by $a(u)$. Feasibility is preserved because $c(a(u)) \le c(u)$, and the objective does not increase; both by (H2).
(T3) Each such replacement strictly decreases the number of identification-dependent elements, because $a(u)$ is identification-free by (H2) and the status of no other element changes. The plan is finite, so the iteration terminates at an identification-free plan $\tilde{S}_\theta$ that is feasible and satisfies $j(\tilde{S}_\theta;\theta) \le j(S_\theta;\theta)$.
(T4) By θ-optimality of S_θ the reverse inequality also holds, so $j(\tilde{S}_\theta;\theta) = j(S_\theta;\theta)$.
(T5) By (H1) the value of any identification-free plan does not depend on θ. Let $j^\star$ be the minimum over identification-free plans, attained at $S^\star$; both are θ-independent.
(T6) Since $\tilde{S}_\theta$ is identification-free, $j(\tilde{S}_\theta;\theta) \ge j^\star$. Since $S^\star$ is feasible, $j^\star = j(S^\star;\theta) \ge j(S_\theta;\theta) = j(\tilde{S}_\theta;\theta)$. The two inequalities force $j(S_\theta;\theta) = j^\star$ for every θ.
(T7) Hence $\mathbb{E}_\theta[j(S_\theta;\theta)] = j^\star$.
(T8) $S^\star$ is available to the prior planner, whose optimum can only be at least as good: $\mathbb{E}_\theta[j(S_\pi;\theta)] \le j^\star$.
(T9) Substituting (T7) and (T8) into (2) gives $\mathrm{EVPI} \le 0$; with $\mathrm{EVPI} \ge 0$ this forces $\mathrm{EVPI} = 0$. □

The argument requires both hypotheses. With (H2) alone, identification-free plans are optimal but their value may still depend on θ, and step (T5) fails. The argument uses nothing specific to expected loss, which gives the following extension.

Corollary 5 (Monotone law-invariant objectives) *Let ρ be a functional of the loss distribution that is monotone with respect to almost sure dominance, and set $J_\rho(S;\theta) = \rho(L(S;\theta,\cdot))$. If (H1) holds and (H2) holds with j replaced by J_ρ, then the value of perfect structural information under J_ρ is zero. This covers expected loss,* CVaR_α *for every α, and spectral risk measures.*

Proof (C1) Steps (T1)–(T4) use only finiteness of the plan set, the cost clause and the dominance inequality, all assumed for J_ρ.
(C2) Step (T5) uses (H1) through the *law* of L: if the law of $L(S;\theta,\cdot)$ does not depend on θ, then neither does $\rho(L(S;\theta,\cdot))$ for any functional ρ of that law. This is where law invariance rather than mean invariance is required.
(C3) Steps (T6)–(T9) are algebraic and carry over with j replaced by J_ρ. □

5 Bounding the value outside the zero-value regime

Theorem 6 (Identification-premium bound, disjoint categories) *Suppose the focal categories' subtrees are node-disjoint, the θ_k are independent across categories, and plans decompose with $c(S) = \sum_k c(S_k)$. For a budget b define*

$$v_k^{\mathrm{inv}}(b) = \min_{c(a)\le b} \mathbb{E}_{\theta_k}\, p_k(a;\theta_k), \tag{5}$$

$$v_k^{\mathrm{full}}(b;\theta_k) = \min_{c(a)\le b} p_k(a;\theta_k), \tag{6}$$

with p_k the starvation probability of focal category k and the minima over prior-available actions. Let $\mathcal{B}_k$ be the finite set of action costs of category k not exceeding the plan budget B. Writing U for the bound, to keep it distinct from that budget,

$$\mathrm{EVPI} \;\le\; U \;:=\; \frac{1}{K_0}\sum_k I_k, \qquad I_k = \mathbb{E}_{\theta_k}\left[\max_{b\in\mathcal{B}_k}\left(v_k^{\mathrm{inv}}(b) - v_k^{\mathrm{full}}(b;\theta_k)\right)^+\right]. \tag{7}$$

Proof (S1) Every plan induces a budget split by setting $\beta_k = c(S_k)$, and every split $\{\beta_k\}$ with $\sum_k \beta_k \le B$, together with per-category actions of cost at most β_k, induces a feasible plan. The correspondence is not one-to-one, since several splits induce the same plan, but that is not needed below.
(S2) By independence of the θ_k and additivity of the loss (1), the prior objective separates across categories once the split is fixed.
(S3) Hence, with $A = \min_S \mathbb{E}_\theta j(S;\theta)$, $K_0 A = \min_{\mathrm{split}} \sum_k v_k^{\mathrm{inv}}(\beta_k)$.
(S4) Similarly, with $M(\theta) = \min_S j(S;\theta)$, $K_0 M(\theta) = \min_{\mathrm{split}} \sum_k v_k^{\mathrm{full}}(\beta_k;\theta_k)$.
(S5) Define the pointwise deficit $d_k(b;\theta_k) = v_k^{\mathrm{inv}}(b) - v_k^{\mathrm{full}}(b;\theta_k)$; it may be negative for particular θ_k, and only its positive part enters.
(S6) Both value functions are nonincreasing step functions of b whose jumps occur only at action costs, so $d_k(\cdot;\theta_k)$ is piecewise constant and $\sup_{b\le B} d_k(b;\theta_k)^+ = \max_{b\in\mathcal{B}_k} d_k(b;\theta_k)^+ =: D_k(\theta_k)$. This is what makes the premium finite to evaluate: a supremum over a continuum of budgets is attained on a finite breakpoint set.
(S7) Pointwise in θ and for every split, $\sum_k v_k^{\mathrm{full}}(\beta_k;\theta_k) \ge \sum_k v_k^{\mathrm{inv}}(\beta_k) - \sum_k D_k(\theta_k)$.
(S8) Minimising both sides over splits preserves the inequality, since $f \ge g - c$ pointwise implies $\min f \ge \min g - c$ for a constant c; the correction term does not depend on the split, which is what allows the minimisation to pass through. Hence $K_0 M(\theta) \ge K_0 A - \sum_k D_k(\theta_k)$.

(S9) Taking $\mathbb{E}_\theta$ of (S8) and using the definition of I_k in (7) gives $K_0\,\mathbb{E}[M] \geq K_0 A - \sum_k I_k$. The slack in the bound is introduced earlier, at (S6) and (S7), where a category's deficit is replaced by its maximum over budgets and the positive part discards the cases in which identification would have hurt.
(S10) Rearranging, EVPI $= A - \mathbb{E}[M] \leq K_0^{-1} \sum_k I_k$. □

Corollary 7 (Dilution) *Call focal category k* isolated *if its subtree contains exactly one latent category (v, k'), the candidates in $C_{v,k'}$ have no inputs and supply no category other than (v, k'), and every other node of the subtree fails with probability zero. Let c_v be the least cost of securing a node through which v's output reaches the focal node, v included, with $c_v = \infty$ if there is none, as when $v = 0$. If focal category k is isolated, with m equal-cost, equal-risk candidates of risk $\bar{q}$, a uniform prior and securing cost c_u, then $I_k = \bar{q}(1 - 1/m)$ if $c_u \leq B$ and $c_u < c_v$, and $I_k = 0$ otherwise.*

Proof (D1) Isolation fixes the starvation probability. With supplier θ, $Z_k = F_\theta$ unless the plan secures θ or a node through which v's output reaches the focal node, in which case $Z_k = 0$: the other candidates feed nothing, and every other node of the subtree is reliable. Hence $p_k(a;\theta) = 0$ if a contains θ or such a node, and $p_k(a;\theta) = \bar{q}$ otherwise. Securing any other node changes nothing.
(D2) If $b < \min\{c_u, c_v\}$ no useful action is affordable, so both value functions equal $\bar{q}$ and the deficit is zero. If $b \geq \min\{c_v, mc_u\}$ both planners can reach zero, by the path cut or by securing every candidate, and the deficit is zero.
(D3) Suppose $c_u < c_v$ and $b \in [jc_u, (j+1)c_u) \cap [c_u, \min\{c_v, mc_u\})$ with $1 \leq j < m$. The informed planner secures the true candidate, so $v^{\text{full}} = 0$. The uninformed planner can afford at most j candidates; under the uniform prior any j of them cover the supplier with probability j/m, so $v^{\text{inv}} = \bar{q}(m-j)/m$. The deficit $\bar{q}(m-j)/m$ is largest at $j = 1$, on $[c_u, \min\{2c_u, c_v\})$, which is nonempty because $c_u < c_v$ and $c_u > 0$.
(D4) That interval contains the breakpoint c_u, which lies in $\mathcal{B}_k$ exactly when $c_u \leq B$. Then $D_k(\theta) = \bar{q}(1 - 1/m)$ for every θ and $I_k = \bar{q}(1 - 1/m)$. If $c_u > B$, every breakpoint in $\mathcal{B}_k$ falls under (D2). If $c_u \geq c_v$, every budget is either below $c_v \leq c_u$ or at least c_v, and again falls under (D2). In both cases $I_k = 0$. □

Isolation is not a formality. In Example 1 of Appendix A, raise the cost of securing w above c_u and keep $c_u \leq B$. Focal category 1 then has two equal-risk candidates with $\bar{q} = 0.1$, but u_1 also supplies (w, b), so the category is not isolated. Its premium is $I_1 = 0.025$, not $\bar{q}(1 - 1/m) = 0.05$: when u_2 is the supplier, securing it still leaves w exposed through u_1, and the informed planner's advantage vanishes.

When the bound is attained.

Corollary 7 gives the premium I_k exactly. It does not say that EVPI $= U$, and in general the two differ. Remark 1 gives one reason, an affordable path cut. A second is that the uninformed planner can move budget between categories. Take two isolated categories, each with $m = 2$ candidates and $c_u = 1$, risks 0.1 and 0.9, unaffordable path cuts and $B = 2$. The uninformed planner secures both candidates of the riskier category, so EVPI $= 0.05$ while $U = 0.25$, although $c_u \leq B < c_v$ in both categories. Equality does hold in the following case. Suppose the assumptions of Theorem 6 hold,

the K_0 focal categories are isolated with a common m, $\bar{q}$ and c_u and a uniform prior, every path cut costs more than B, and $K_0 c_u \leq B < (K_0+1)c_u$. Then EVPI $= U = \bar{q}(1-1/m)$.
(E1) Every category has $c_u \leq B < c_v$, so Corollary 7 gives $I_k = \bar{q}(1-1/m)$ for each k, and $U = \bar{q}(1-1/m)$.
(E2) Securing the K_0 true suppliers costs $K_0 c_u \leq B$ and sets every Z_k to zero, so $\min_S j(S;\theta) = 0$ for every θ.
(E3) A plan the uninformed planner can afford contains no path cut and at most $\lfloor B/c_u \rfloor = K_0$ candidates, and securing any other node of the subtrees has no effect, by (D1). If it secures n_k candidates of category k, that category starves with prior probability $\bar{q}(m-n_k)^+/m$. Since $\sum_k n_k \leq K_0$, K_0 times its objective is at least $\bar{q}\sum_k (m-n_k)/m \geq \bar{q}K_0(m-1)/m$, with equality at $n_k = 1$ for every k.
(E4) Hence the prior-optimal value is $\bar{q}(1-1/m)$, and by (E2) EVPI $= \bar{q}(1-1/m) = U$.

Proposition 8 (A checkable witness for (H2)) *Fix a latent category (v,k) with $v \neq 0$, so that v is itself securable, and suppose that (i) every candidate $u \in C_{v,k}$ has v as its only consumer, and (ii) $c_v \leq \min_{u \in C_{v,k}} c_u$. Then securing v is an identification-free replacement satisfying both clauses of (H2) for every $u \in C_{v,k}$. Both conditions are functions of the public graph and the public cost vector, so they can be checked before any structural discovery is commissioned.*

Proof (W1) Securing v guarantees v's output to its consumers in every scenario and for every $\theta_{v,k}$, so under any plan containing that action the focal loss is a function of the scenario and of the plan's other elements alone, and its conditional law does not depend on $\theta_{v,k}$: the action is identification-free in the sense of Definition 1. The service of (v,k) itself is unchanged; what the action removes is its influence on L.
(W2) The cost clause is (ii) directly: $c_v \leq c_u$ for every candidate $u \in C_{v,k}$.
(W3) For the dominance clause, fix θ and a plan S containing u. Under the conjunctive cascade, securing u affects the focal loss only through the categories u serves, and by (i) the only such category is (v,k); its effect there is to remove u's own failure from that category, which helps only when u is the true supplier and would otherwise fail. Securing v instead guarantees v's output outright, which removes the failure of (v,k) and of every other category v consumes, for every θ and every scenario.
(W4) Hence the operability of every node downstream of v under $S \setminus \{u\} \cup \{v\}$ is at least that under S, scenario by scenario, so $L(S \setminus \{u\} \cup \{v\}; \theta, \omega) \leq L(S;\theta,\omega)$ pointwise, and therefore $j(S \setminus \{u\} \cup \{v\}; \theta) \leq j(S;\theta)$ for every θ. With (W1) and (W2), (H2) holds with $a(u) = v$. □

The hypothesis $v \neq 0$ is not cosmetic. Plans are subsets of $V \setminus \{0\}$, so the focal node cannot be secured and the witness is unavailable for exactly the K_0 categories that enter the loss directly. Proposition 8 closes the gap left by (H2) in one direction only. It gives a sufficient condition a firm can test on its own graph and cost data; it does not decide (H2) in general, and its condition (i) fails wherever a candidate supplies more than one consumer. Section 7 records what remains open.

Remark 1 (The premium is loose where the theorem bites) The maximum in (7) ranges over all breakpoints, including budgets a planner would not choose. If $c_v \leq B$, so that the consumer cut is affordable, both planners take it and reach zero loss for that category, so its

contribution to EVPI is zero; this follows directly, not through Theorem 4, whose cost clause does not hold when $c_u < c_v$. Yet Corollary 7 still returns $I_k = \bar{q}(1 - 1/m) > 0$ in exactly that case. The bound remains valid, being an upper bound, but it is not tight there, and a positive premium therefore does not show that identifying the channel has any decision value. This accounts for a substantial part of the mass near zero in Figure 1: among the enumerated generic instances in which every category has an affordable cut, the mean tightness ratio is 0.102 and 67% of instances fall below 0.05, against 0.203 and 41% among those in which no category does. The looseness is a property of the bound itself.

Corollary 9 (Per-channel bound) *Under the assumptions of Theorem 6, the partial value of learning θ_k alone satisfies* $\mathrm{EVPPI}_k \le I_k/K_0$, *where* $\mathrm{EVPPI}_k = A - \mathbb{E}_{\theta_k}\big[\min_S \mathbb{E}_{\theta_{-k}} j(S;\theta)\big]$.

Proof (P1) With θ_k known and the remaining components unknown, the separation of step (S2) gives, for each split, $K_0\,\mathbb{E}_{\theta_{-k}} j = v_k^{\mathrm{full}}(\beta_k;\theta_k) + \sum_{j\neq k} v_j^{\mathrm{inv}}(\beta_j)$.
(P2) By (S6), $v_k^{\mathrm{full}}(\beta_k;\theta_k) \ge v_k^{\mathrm{inv}}(\beta_k) - D_k(\theta_k)$ for every β_k.
(P3) Adding $\sum_{j\neq k} v_j^{\mathrm{inv}}(\beta_j)$ to both sides of (P2) and minimising over splits, permissible because $D_k(\theta_k)$ does not depend on the split, gives $K_0 \min_S \mathbb{E}_{\theta_{-k}} j \ge K_0 A - D_k(\theta_k)$.
(P4) Taking $\mathbb{E}_{\theta_k}$ and rearranging gives $\mathrm{EVPPI}_k \le I_k/K_0$. □

5.1 Dispersion under a specified construction

Heterogeneous latent candidate risk can be introduced through a dispersion parameter λ. Monotonicity in a vague notion of "more dispersion" would not be well posed, since it leaves open which ordering of distributions is meant, so we fix the family and compute the premium exactly.

Definition 2 (Two-point dispersion family $\mathcal{D}(\lambda)$) A latent category has m candidates and a uniform prior over which of them is the supplier. Candidate risks take the two values $\bar{q} \pm \lambda\delta$ with equal probability, independently of which candidate is the supplier, where $\lambda \in [0,1]$, $\delta > 0$ and $\bar{q} \pm \delta \in [0,1]$. The assignment of risks to candidates is latent and exchangeable under the prior, and is revealed only by probing. The category is an isolated focal category in the sense of Corollary 7, and c_v is the least cost of a path cut as defined there. Actions are: no action at cost 0; secure a named candidate at cost c_u; secure the consumer at cost $c_v > c_u$.

Proposition 10 (Premium under $\mathcal{D}(\lambda)$) *Write $D = \bar{q}(1 - 1/m)$ for the dilution term of Corollary 7. Under $\mathcal{D}(\lambda)$,*

$$I_k(\lambda) = \begin{cases} \tfrac{1}{2}\max\{\lambda\delta,\, D\} + \tfrac{1}{2}D, & c_u \le B, \\ \tfrac{1}{2}\lambda\delta, & c_u > B. \end{cases} \tag{8}$$

In both cases I_k is nondecreasing in λ. In the first it is constant for $\lambda \le D/\delta$ and strictly increasing above that threshold; in the second it is strictly increasing throughout. If the risk assignment is instead public, the prior planner can target the fragile candidate directly and the premium need not increase in λ.

Proof (R1) At $b = 0$ the only available action is inaction, so $v^{\text{inv}}(0) = \mathbb{E}[q_\theta] = \bar{q}$ by the mean-preserving construction, while $v^{\text{full}}(0;\theta) = q_\theta$. The deficit is $\bar{q} - q_\theta$, whose positive part equals $\lambda\delta$ when the supplier carries the low risk and 0 otherwise, each with probability $\frac{1}{2}$.
(R2) For $b \in [c_u, c_v)$ the informed planner secures the supplier, so $v^{\text{full}} = 0$. The uninformed planner faces exchangeable candidates — this is where latency of the assignment is used — so every named candidate has the same prior value; securing one leaves the category exposed with probability $(m-1)/m$ at conditional risk $\bar{q}$, giving $v^{\text{inv}} = D$. This is below the do-nothing value $\bar{q}$, so it is the prior planner's best action on that interval. The deficit is D for both risk types.
(R3) For $b \geq c_v$ both planners cut the consumer and reach zero, so the deficit is zero.
(R4) If $c_u \leq B$, the maximum over breakpoints is $\max\{\lambda\delta, D\}$ when the supplier carries the low risk, and D when it carries the high risk. Averaging over the two equally likely types gives the first line of (8).
(R5) If $c_u > B$, only the breakpoint $b = 0$ is feasible and (R1) gives the second line.
(R6) Both expressions are nondecreasing in λ because $\lambda \mapsto \max\{\lambda\delta, D\}$ and $\lambda \mapsto \lambda\delta$ are nondecreasing; the threshold statement follows from $\lambda\delta \geq D \iff \lambda \geq D/\delta$.
(R7) If the assignment is public, the prior planner may secure the high-risk candidate deliberately, which changes v^{inv} and removes the exchangeability used in (R2), so monotonicity no longer follows from this argument. □

Equation (8) predicts a threshold rather than a slope. Below $\lambda = D/\delta$ the dilution term dominates and dispersion contributes nothing to the premium; only above it does dispersion matter.

5.2 Why the bound does not transfer to CVaR

Theorem 4 extends to CVaR at no cost (Corollary 5). Theorem 6 does not, and the obstruction is structural.

Proposition 11 (What can be said) *Let $L \in [0,1]$. Then $\mathbb{E}[L] \leq \text{CVaR}_\alpha(L) \leq \mathbb{E}[L]/(1-\alpha)$, and, writing $\text{EVPI}_{\text{CVaR}}$ for the value of perfect information under the CVaR objective,*

$$\text{EVPI}_{\text{CVaR}} \;\leq\; \frac{1}{1-\alpha}\,\text{EVPI} \;+\; \frac{\alpha}{1-\alpha}\,\mathbb{E}_\theta\left[\min_S j(S;\theta)\right]. \tag{9}$$

Proof (V1) By the Rockafellar–Uryasev representation [23], $\text{CVaR}_\alpha(L) = \min_t\{t + (1-\alpha)^{-1}\mathbb{E}(L-t)^+\}$.
(V2) Evaluating at $t = 0$ and using $L \geq 0$ gives $\text{CVaR}_\alpha(L) \leq \mathbb{E}[L]/(1-\alpha)$.
(V3) For any t, since $(1-\alpha)^{-1} \geq 1$ and $\mathbb{E}(L-t)^+ \geq 0$, $t + (1-\alpha)^{-1}\mathbb{E}(L-t)^+ \geq t + \mathbb{E}(L-t)^+ \geq t + \mathbb{E}(L-t) = \mathbb{E}[L]$; minimising over t gives $\text{CVaR}_\alpha(L) \geq \mathbb{E}[L]$.
(V4) With $A_{\text{CVaR}} = \min_S \mathbb{E}_\theta J(S;\theta)$, (V2) gives $A_{\text{CVaR}} \leq (1-\alpha)^{-1}A$.
(V5) By (V3), $\mathbb{E}_\theta \min_S J(S;\theta) \geq \mathbb{E}_\theta \min_S j(S;\theta) = \mathbb{E}[M]$.
(V6) Subtracting and using $A = \text{EVPI} + \mathbb{E}[M]$ yields (9). □

Inequality (9) is not a multiplicative transfer. Its second term is proportional to the residual loss $\mathbb{E}_\theta[\min_S j(S;\theta)]$, the expected loss that remains after optimal mitigation, which is of the same order as the objective itself. At $\alpha = 0.9$ the coefficient $\alpha/(1-\alpha)$ is 9, so that term is nine times a quantity of the order of the objective, and the inequality

is too weak to be useful. We therefore do not use it. The obstruction is visible at step (S2) of Theorem 6, which requires the objective to be *additive* across categories. CVaR is subadditive, and subadditivity gives an inequality, the second statement of Proposition 3, in the direction that destroys the decomposition. A CVaR version of Theorem 6 would need a different argument, which we do not have.

5.3 Unresolved extension: shared subtrees

When category subtrees share nodes, step (S2) fails and the decomposition fails with it. We state what we believe and have not proved.

Conjecture 12 *Under a single-cut regime, meaning that optimal plans take at most one action per focal-category channel, and with replacement slack, the bound (7) holds when each shared channel's premium is counted once per focal category it feeds.*

Section 6 reports an enumeration search for counterexamples that found none, which is evidence about a finite class of small instances and not a proof.

Table 2 Assumptions, results and where each applies.

result	required assumptions	establishes	applicability
Lem. 2	finite feasible plan set	acquisition value $\leq$ EVPI	general
Prop. 1	implemented planner defined in both regimes	$H = \text{EVPI} + \varepsilon_\pi - \varepsilon_\theta$	general
Thm. 4	(H1), (H2)	$\text{EVPI} = 0$	(H1) via (H1′), checkable in advance; (H2) via Prop. 8 or a plan-space argument
Prop. 8	$v \neq 0$, candidates sole-consumed by v, $c_v \leq \min_u c_u$	(H2) holds for that category	checkable from public data
Cor. 9	as Thm. 6	$\text{EVPPI}_k \leq I_k / K_0$	per category
Cor. 5	(H1), (H2) for a monotone law-invariant objective	zero value under CVaR	theoretical
Thm. 6	disjoint subtrees, independent θ_k, additive budget	$\text{EVPI} \leq U$	subset of networks
Cor. 7	isolated category, equal cost and risk, uniform prior	exact premium I_k; EVPI $= U$ only in a sub-case	special case
Prop. 10	family $\mathcal{D}(\lambda)$	premium in closed form	specified family
Conj. 12	shared channels, single cut, slack	unproved bound	exploratory

Table 2 collects the assumptions each result needs and where it applies, since the hypotheses differ from result to result and the distinctions matter for what may be claimed in any application.

6 Computational verification

The experiments in this section are not offered as proof of the analytical results. They serve two purposes: verifying that the implementation matches the definitions, and searching finite instances for violations that would reveal errors in the stated results. In each category a plan may take no action, cut at the consumer, or secure any nonempty subset of the candidates. Every instance is enumerated exactly, over all plans and all θ, with no sampling inside an instance; seeds are used only to draw the instances themselves, and the seeds are listed in the statement on data and code availability.

Table 3 Enumeration checks. A single violation would falsify the corresponding statement.

batch	inst.	viol.	auxiliary check	tightness
generic random	6000	0	—	mean 0.132, max 0.985
Thm. 4 regime	2000	0	nonzero-EVPI cases: 0	—
Cor. 7 regime	2000	0	formula mismatches: 0	mean 0.212, max 1.000

Table 3 reports the three batches, and Figure 1 shows the distribution of the ratio of exact EVPI to the bound. Most instances lie near zero, so the bound is loose in general. The dilution batch is loose too, with mean ratio 0.212: the premium is exact there, but the bound is not. Equality is reached only on some of its instances, and it holds exactly in the sub-case identified after Corollary 7, so the bound cannot be multiplied by any constant smaller than one without further assumptions.

The premium against exact EVPPI.

On miniature disjoint instances we computed the exact per-channel EVPPI_k by enumeration and compared it with the quantity Corollary 9 says bounds it, namely I_k/K_0. Both sides carry the $1/K_0$ factor, so the slack reported here is slack in the bound and not the divisor. Over five independent seeds and 24,981 channels, I_k/K_0 never fell below the enumerated EVPPI_k, and the whole-instance bound was never violated. On 25.6% of channels EVPPI_k is zero while the premium I_k/K_0 is positive, so on those channels the bound carries no information about the value of identification. Where EVPPI_k is strictly positive, the premium is a median 1.94 times it, with interquartile range 1.37 to 3.50. Over all channels the median absolute gap $I_k/K_0 - \mathrm{EVPPI}_k$ is 0.023, with interquartile range 0.011 to 0.040. Rank agreement is moderate and stable across seeds (Spearman 0.501 to 0.518; Pearson 0.573 to 0.589). The rank statistic is computed after merging values within 10^{-12} of each other, because on about a quarter of channels, the 25.6% above, EVPPI_k is mathematically zero but stored with floating-point residue, which would otherwise make its fourth decimal arbitrary.

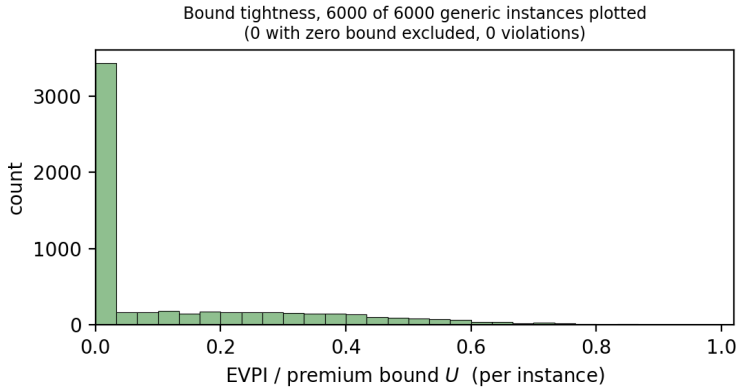


Fig. 1 Ratio of exact EVPI to the bound of Theorem 6 for all 6000 enumerated generic instances, every one of which is plotted. Ratios within 10^{-12} of zero, floating-point residues of an exact zero, are plotted as zero. No value exceeds one.

Search for a counterexample to Conjecture 12.

Over five seeds and 100,000 instances in which channels feed several focal categories, the conjectured bound was never violated and the minimum slack was zero, so it is attained with equality on some instances. The search covers instances with at most three channels and three focal categories; a counterexample may exist outside that range.

7 Scope and limitations

(1) The conditions and the bound are not evaluated on real supply networks here. Every result is either proved or checked by exact enumeration on synthetic miniature instances. Whether (H1′) or (H2) holds on any observed network, how large the identification premiums are there, and how close the bound comes to EVPI at realistic scale are open questions. Answering them on observed multi-tier structure, with the latent layers it actually has, is the natural next step and is left to future work.

(2) The verification is evidence about a finite class of small instances, with at most three focal categories and three candidates per category, and is not proof. A violation outside that class is not excluded by it.

(3) Theorem 6 requires node-disjoint categories. Real networks typically share upstream nodes, and Conjecture 12, which would cover them, is open; the enumeration search supports it only on instances with at most three channels and three focal categories. The CVaR version of Theorem 6 does not follow from the expected-loss version, and Proposition 11 is too weak to substitute for it.

(4) The bound can be loose. Remark 1 identifies one source, an affordable consumer cut, and on the generic enumerated instances the mean ratio of EVPI to the bound is 0.132. It is attained with equality only in the sub-case identified after Corollary 7.

(5) (H2) has no general verification procedure. Proposition 8 is one sufficient route; its condition (i) fails wherever a candidate supplies more than one consumer, and in the family of Definition 2 the cost ordering makes (H2) fail.

(6) A firm that runs a heuristic planner with and without θ observes H, not EVPI. Proposition 1 relates the two by an identity whose sign is undetermined, and we give no bound on the gap between them.

(7) All results concern one mitigation problem: budgeted backup qualification under a conjunctive cascade. Inventory buffering, dual sourcing and contractual remedies may value structural information differently. Concurrent dual sourcing in particular is outside the model, as Section 3 notes: the candidate set is epistemic, and no category is served by two suppliers at once.

8 Conclusion

When the candidates of every supply category are exchangeable by a symmetry of the public network that carries their consumers, upstream subtrees and correlation clusters, *and* identification-free actions of no greater cost uniformly dominate, the decision value of hidden supply-network structure is exactly zero, and by Lemma 2 no information-acquisition policy can recover it. The statement concerns the decision problem rather than any method, and it holds for any monotone law-invariant objective, including CVaR. Where those conditions fail, the value is bounded above, for node-disjoint categories, by identification premiums computable in advance from prior and public data, and the bound is attained in a sub-case of a dilution special case. Exact enumeration of synthetic miniature instances found no violation of any of these statements and no counterexample to the conjectured extension to shared subtrees; it also shows the bound to be loose on typical instances. None of this has been evaluated on a real supply network.

The practical reading is that the value can be examined in advance. Before commissioning sub-tier discovery, a firm can ask whether its situation satisfies conditions under which the answer is provably nothing, and if not, what the largest possible payoff would be. Verifying (H1) through (H1′) is a search for automorphisms of the public structure, not a comparison of the candidates' marginal risks, costs or prior weights, and it is global: a single category whose candidates differ in their consumers, upstream subtrees or clusters defeats it. Verifying (H2) requires an argument over the plan space, for which we give no general procedure, and its cost clause is a real restriction rather than a formality. Within the model there is one route to it: a consumer-level securing action that is no dearer than securing a named candidate. Proposition 8 makes that route checkable from public data when, in addition, the consumer is not the focal node and every candidate has it as its only consumer. The cost condition is exactly what Definition 2 makes false: there the consumer cut costs $c_v > c_u$ and securing every candidate costs mc_u, so both canonical identification-free replacements are dearer than the identification-dependent action they replace, and (H2) fails. We have no second in-model witness. Dual-sourcing or buffering a shared downstream stage would serve, but the model has no inventory and no concurrently multi-sourced category (Section 3); a contractual remedy priced per category rather than per supplier would also serve, but

it is not an element of V. Either would require extending the model. The enumerated batch of Table 3 instantiates (H1) and (H2) by construction at miniature scale, by drawing the consumer cut below the candidate cost. No real network is shown here to satisfy them. Where no such action exists, the theorem is silent and Theorem 6 is the relevant statement.

Declarations

Competing interests.

The author declares no competing interests.

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Data and code availability

No external data are used. Every reported number comes from exact enumeration of synthetic miniature instances drawn from fixed seeds, computed with Python 3.12, NumPy 1.26.4, SciPy 1.11.4 and pandas 2.2.3. Three scripts produce them: `verify_theorem.py` (Table 3 and Figure 1; seed 20260810), `exp_A_idx_vs_evppi.py` (the comparison with exact EVPPI; 2000 instances at each of the seeds 20260812, 733, 40427, 991 and 20260101) and `exp_E_conjecture1.py` (the search on Conjecture 12; 20000 instances at each of the same seeds). The figures in Remark 1 are computed from the instances of `verify_theorem.py` by a short additional analysis.

What can be run today is less than this. The public repository https://github.com/rahul-k-mishra/supply-network-information-value, at commit `9e4b23b`, holds a version of `verify_theorem.py` that reproduces Table 3 but plots Figure 1 without the treatment of floating-point residues described in its caption, and earlier versions of the other two scripts, which use a narrower action set and an unmerged rank statistic and so do not reproduce the numbers reported in Section 6. The versions of all three scripts used for this paper, the analysis behind Remark 1, the console output of every reported run and a specification of the environment will be deposited in that repository.

Appendix A The sufficient condition for (H1)

Throughout, F_v is the failure indicator of node v, operability is defined recursively by $O_v = 1$ if $v \in S$ and $O_v = (1 - F_v) \prod_{k \in K_v} O_{s_{v,k}(\theta)}$ otherwise, where $s_{v,k}(\theta)$ is $\theta_{v,k}$ for a latent category and the public supplier otherwise, and $Z_k = 1 - O_{s_{0,k}(\theta)}$.

Example 1 (Marginal exchangeability is not sufficient) The focal node takes category 1 from w and category 2 from x, both public, so $K_0 = 2$. Node w consumes a latent category a with $C_{w,a} = \{u_1, u_2\}$ and a public category b supplied by u_1; x, u_1 and u_2 have no inputs. Failures are independent, with $q_{u_1} = q_{u_2} = 0.1$, $q_x = 0.15$ and $q_w = 0$. All securing costs

equal 1, the budget is $B = 1$, and the prior is uniform. The candidates have the same failure law, cost and prior weight, and differ only in that u_1 also supplies (w, b).
(X1) Under the empty plan $Z_2 = F_x$. If $\theta = u_1$ then $O_w = 1 - F_{u_1}$, so $\mathbb{P}(Z_1 = 1) = 0.1$ and L takes the values $0, \frac{1}{2}, 1$ with probabilities $0.765, 0.220, 0.015$. If $\theta = u_2$ then $O_w = (1 - F_{u_1})(1 - F_{u_2})$, so $\mathbb{P}(Z_1 = 1) = 0.19$ and the probabilities are $0.6885, 0.2830, 0.0285$. The law of L, and even its mean (0.125 against 0.170), depends on θ, so (H1) fails.
(X2) Proposition 8 applies to (w, a): $w \neq 0$, w is the only consumer of u_1 and of u_2, and $c_w \leq \min_u c_u$. Securing w sets $Z_1 = 0$ in every scenario, so it dominates securing either candidate pointwise at no greater cost, and (H2) holds.
(X3) The feasible plans are $\emptyset, \{w\}, \{x\}, \{u_1\}, \{u_2\}$, with $j(\cdot; u_1) = 0.125, 0.075, 0.050, 0.075, 0.125$ and $j(\cdot; u_2) = 0.170, 0.075, 0.095, 0.125, 0.125$. The state-optimal values are 0.050 and 0.075, with mean 0.0625; the prior-optimal plan is $\{x\}$, with value 0.0725. Hence EVPI $= 0.01 > 0$. Step (T5) of Theorem 4 fails: both state-optimal plans are identification-free, but $j(\{x\}; \theta)$ varies with θ.

Variants in which u_1 alone has an upstream supplier that also feeds x, or in which u_1 alone shares a copula cluster with x, keep the mean of L fixed and change its law, so they defeat Corollary 5 while leaving expected loss unchanged.

Lemma 13 (Sufficiency of (H1′)) *Let σ be a structure automorphism. For every plan S and state θ, $Z(\sigma S; \sigma\theta, \cdot)$ has the same law as $\Pi_\sigma Z(S; \theta, \cdot)$, where Π_σ permutes the focal coordinates by σ_0. Consequently (H1′) implies (H1), and the local form stated in Section 4 implies (H1′).*

Proof (Q1) Fix a failure vector F and define G by $G_{\sigma v} = F_v$. The map σ carries the supplier graph of θ onto that of $\sigma\theta$: for a public category this is (A1), and for a latent one $s_{\sigma v, \sigma_v k}(\sigma\theta) = \sigma(\theta_{v,k}) = \sigma(s_{v,k}(\theta))$ by the definition of $\sigma\theta$. Hence σv has the same depth in the graph of $\sigma\theta$ as v in the graph of θ.
(Q2) We show $O_{\sigma v}(\sigma S, \sigma\theta, G) = O_v(S, \theta, F)$ by induction on that depth. Membership agrees because σ is a bijection; own failure agrees because $G_{\sigma v} = F_v$; and by (Q1) and $K_{\sigma v} = \sigma_v(K_v)$ the inputs of σv are the images $\sigma(s_{v,k}(\theta))$, $k \in K_v$, which lie at smaller depth, so the induction hypothesis equates their operability with that of $s_{v,k}(\theta)$. Substituting into the recursion gives the claim; at depth zero the product is empty.
(Q3) By (A0) and (Q1) the supplier of focal category $\sigma_0 k$ under $\sigma\theta$ is $\sigma(s_{0,k}(\theta))$, so (Q2) gives $Z_{\sigma_0 k}(\sigma S, \sigma\theta, G) = Z_k(S, \theta, F)$ for every F.
(Q4) Under the one-factor Gaussian copula, $F_v = \mathbf{1}\{\sqrt{r_g}\, Y_g + \sqrt{1 - r_g}\, \varepsilon_v \leq \Phi^{-1}(q_v)\}$ with $g = \kappa(v)$, r_g the cluster's copula parameter, and (Y_g), (ε_v) independent standard normal. By (A3), $G_v = F_{\sigma^{-1} v}$ is the same function of $Y'_g = Y_{\eta^{-1} g}$ and $\varepsilon'_v = \varepsilon_{\sigma^{-1} v}$ that F_v is of Y and ε. These are re-indexings of independent standard normal families by bijections, so G has the law of F. With (Q3) this gives the first statement.
(Q5) Given an identification-free S and θ, θ', take σ from (H1′). Since $\sigma S = S$ and $\sigma\theta = \theta'$, (Q4) gives $Z(S; \theta', \cdot) \overset{d}{=} \Pi_\sigma Z(S; \theta, \cdot)$. The loss (1) is symmetric in the focal coordinates, so $L(S; \theta', \cdot) \overset{d}{=} L(S; \theta, \cdot)$, which is (H1).
(Q6) For the local form, list the latent categories $\ell_1, \ldots, \ell_n$ and set $\theta^{(0)} = \theta$. At step i apply the automorphism τ_i that maps $\theta^{(i-1)}_{\ell_i}$ to θ'_{ℓ_i} (the identity if they agree). It fixes ℓ_i, so the ℓ_i coordinate becomes θ'_{ℓ_i}; it fixes every other latent category together with its candidates, so the other coordinates are unchanged. After n steps the state is θ'. The composition $\tau_n \circ \cdots \circ \tau_1$ satisfies (A0)–(A3), maps θ to θ' and fixes S, which is (H1′). □

The requirement $\sigma S = S$ in (H1′) may be weakened to $L(\sigma S; \theta', \cdot) \overset{d}{=} L(S; \theta', \cdot)$ with the same proof, which matters when an identification-free plan meets two candidates' upstream cones asymmetrically without changing the loss. (H1′) is sufficient, not necessary: (H1) can hold by numerical coincidence without any symmetry. Steps (T5) of Theorem 4 and (C2) of Corollary 5 use only (H1), so both results hold unchanged with (H1′) in place of (H1).

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